

development

April 20, 2026

1 Baby GPT — Developmental Analysis

Tracks how the model develops over longer training horizons (5m $\rightarrow$ 15m $\rightarrow$ 30m $\rightarrow$ 1h $\rightarrow$ 2h $\rightarrow$ 4h $\rightarrow$ 8h $\rightarrow$ 12h).

Milestone eval + prompt-pack generation now happens **inside** `train.py` (see `_save_timed_checkpoint`), so no post-run CLI steps are needed — the notebook reads `model_<label>.json` sidecars and `progress.csv` directly from the latest run dir.

A prior-run overlay is supported via `BASELINE_DIR` in the next cell.

```
[1]: import json
import os
import re
import math
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import numpy as np
from pathlib import Path
from IPython.display import display, Markdown

HERE = Path(".").resolve()
OUT = HERE / "output"

def _looks_like_run_dir(d: Path) -> bool:
    """A run dir contains training artifacts (checkpoint sidecars, progress.csv,
    or eval_results.csv). Excludes archive dirs and non-run scratch like_
    ↪ exports/."""
    if d.name.startswith("_"):
        return False
    if (d / "progress.csv").exists() or (d / "eval_results.csv").exists():
        return True
    return any(d.glob("model_*.json"))

def _latest_run_dir():
    """Return the most recent run-dir subdir under output/.
```

```

    Sorted by mtime (most recently written run wins) with name as tiebreaker."
    if not OUT.is_dir():
        return OUT
    candidates = [d for d in OUT.iterdir() if d.is_dir() and
↪ _looks_like_run_dir(d)]
    if not candidates:
        return OUT
    candidates.sort(key=lambda d: (d.stat().st_mtime, d.name), reverse=True)
    return candidates[0]

RUN_DIR = _latest_run_dir()
print(f"RUN_DIR: {RUN_DIR}")

# Optional baseline run to overlay against (yesterday's dedicated 5m-4h sweep).
BASELINE_DIR = OUT / "2026-04-17_phase2_baseline"
if not BASELINE_DIR.is_dir() or BASELINE_DIR == RUN_DIR:
    BASELINE_DIR = None
print(f"BASELINE: {BASELINE_DIR}")

LABELS = ["5m", "15m", "30m", "1h", "2h", "4h", "8h", "12h"]
LABEL_S = {
    "5m": 300,
    "15m": 900,
    "30m": 1800,
    "1h": 3600,
    "2h": 7200,
    "4h": 14400,
    "8h": 28800,
    "12h": 43200,
}
COLORS = [
    "#4e79a7",
    "#f28e2b",
    "#e15759",
    "#76b7b2",
    "#59a14f",
    "#b07aa1",
    "#ff9da7",
    "#9c755f",
]
COLOR_OF = {l: c for l, c in zip(LABELS, COLORS)}

def _eval_df_from_dir(run_dir):
    """Build the (label, elapsed_seconds, num_steps, train_loss_ema, val_bpb)
↪ table

```

```

from JSON sidecars. Falls back to eval_results.csv if present and non-empty.
↪ """
    if run_dir is None or not Path(run_dir).is_dir():
        return pd.DataFrame()
    csv = Path(run_dir) / "eval_results.csv"
    if csv.exists():
        try:
            df = pd.read_csv(csv)
            if len(df) > 0:
                return df
        except Exception:
            pass
    rows = []
    for label in LABELS:
        meta = Path(run_dir) / f"model_{label}.json"
        if not meta.exists():
            continue
        try:
            d = json.loads(meta.read_text())
        except Exception:
            continue
        rows.append(
            {
                "label": label,
                "elapsed_seconds": d.get("elapsed_seconds"),
                "num_steps": d.get("num_steps"),
                "train_loss_ema": d.get("train_loss_ema"),
                "val_bpb": d.get("val_bpb"),
            }
        )
    return pd.DataFrame(rows)

plt.rcParams.update(
    {"figure.dpi": 130, "axes.spines.top": False, "axes.spines.right": False}
)

```

RUN_DIR: C:\Dev\AI\Training\autoresearch\output\2026-04-19_2h_warmdown_0.90_run
 BASELINE: C:\Dev\AI\Training\autoresearch\output\2026-04-17_phase2_baseline

1.1 1. Checkpoint Inventory

```

[2]: # Load JSON sidecars for all found checkpoints
rows = []
for label in LABELS:
    pt = RUN_DIR / f"model_{label}.pt"
    meta = RUN_DIR / f"model_{label}.json"

```

```

    if not pt.exists():
        continue
    d = {"label": label, "pt_size_mb": round(pt.stat().st_size / 1e6, 1)}
    if meta.exists():
        d.update(json.loads(meta.read_text()))
    rows.append(d)

inventory = pd.DataFrame(rows)
print(f"Found {len(inventory)} checkpoint(s)")
display(
    inventory[
        [
            c
            for c in [
                "label",
                "elapsed_seconds",
                "num_steps",
                "train_loss_ema",
                "val_bpb",
                "pt_size_mb",
            ]
            if c in inventory.columns
        ]
    ]
)

```

Found 5 checkpoint(s)

	label	elapsed_seconds	num_steps	train_loss_ema	val_bpb	pt_size_mb
0	5m	300.1	1443	3.512730	1.253732	90.2
1	15m	900.1	4331	3.378381	1.205067	90.2
2	30m	1800.1	8664	3.304161	1.177505	90.2
3	1h	3600.1	17328	3.223759	1.142390	90.2
4	2h	7200.2	34661	2.948463	1.050018	90.2

1.2 2. Developmental Curve — val_bpb & train loss vs time

```

[3]: # Build eval_df from JSON sidecars (or eval_results.csv if present).
eval_df = _eval_df_from_dir(RUN_DIR)
if eval_df.empty:
    print("No milestone checkpoints found in RUN_DIR yet.")
else:
    eval_df["elapsed_min"] = (
        pd.to_numeric(eval_df["elapsed_seconds"], errors="coerce") / 60
    )
    eval_df["val_bpb"] = pd.to_numeric(eval_df["val_bpb"], errors="coerce")
    eval_df["train_loss_ema"] = pd.to_numeric(
        eval_df["train_loss_ema"], errors="coerce"
    )

```

```

    )
    display(eval_df)

# Optional dense periodic curve (progress.csv) written by train.py every ~5% of
↳ progress.
progress_csv = RUN_DIR / "progress.csv"
progress_df = pd.read_csv(progress_csv) if progress_csv.exists() else pd.
↳ DataFrame()
if not progress_df.empty:
    progress_df["elapsed_min"] = (
        pd.to_numeric(progress_df["elapsed_seconds"], errors="coerce") / 60
    )
    progress_df["val_bpb_capped"] = pd.to_numeric(
        progress_df["val_bpb_capped"], errors="coerce"
    )

# Optional baseline for side-by-side overlay.
baseline_df = _eval_df_from_dir(BASELINE_DIR)
if not baseline_df.empty:
    baseline_df["elapsed_min"] = (
        pd.to_numeric(baseline_df["elapsed_seconds"], errors="coerce") / 60
    )
    baseline_df["val_bpb"] = pd.to_numeric(baseline_df["val_bpb"],
↳ errors="coerce")
    baseline_df["train_loss_ema"] = pd.to_numeric(
        baseline_df["train_loss_ema"], errors="coerce"
    )

if not eval_df.empty:
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))

    # --- val_bpb ---
    ax = axes[0]
    # Dense periodic curve (capped eval) behind the milestone markers
    if not progress_df.empty:
        ax.plot(
            progress_df["elapsed_min"],
            progress_df["val_bpb_capped"],
            linewidth=1,
            color="#4e79a7",
            alpha=0.35,
            label="periodic (capped eval)",
        )
    # Optional baseline curve for comparison
    if not baseline_df.empty:
        ax.plot(
            baseline_df["elapsed_min"],

```

```

        baseline_df["val_bpb"],
        marker="s",
        linestyle="--",
        linewidth=1.5,
        color="#888",
        markersize=6,
        alpha=0.7,
        label="baseline (2026-04-17)",
    )
ax.plot(
    eval_df["elapsed_min"],
    eval_df["val_bpb"],
    marker="o",
    linewidth=2,
    color="#4e79a7",
    markersize=8,
    label="milestones (full eval)",
)
for _, row in eval_df.iterrows():
    if pd.notna(row["val_bpb"]):
        ax.annotate(
            f"{row['val_bpb']:.4f}",
            (row["elapsed_min"], row["val_bpb"]),
            textcoords="offset points",
            xytext=(6, 6),
            fontsize=9,
        )
ax.set_xlabel("Elapsed training time (min)")
ax.set_ylabel("val_bpb (lower is better)")
ax.set_title("Validation BPB vs Training Time")
ax.grid(True, alpha=0.2)
ax.legend(loc="upper right", fontsize=9)

# Marginal improvement annotation
bpb_vals = eval_df["val_bpb"].dropna().values
if len(bpb_vals) >= 2:
    total_delta = bpb_vals[0] - bpb_vals[-1]
    ax.set_title(f"Validation BPB vs Training Time ( $\Delta$ ={{total_delta:+.4f}}_
↳total)")

# --- train loss EMA ---
ax2 = axes[1]
ax2.plot(
    eval_df["elapsed_min"],
    eval_df["train_loss_ema"],
    marker="s",
    linewidth=2,

```

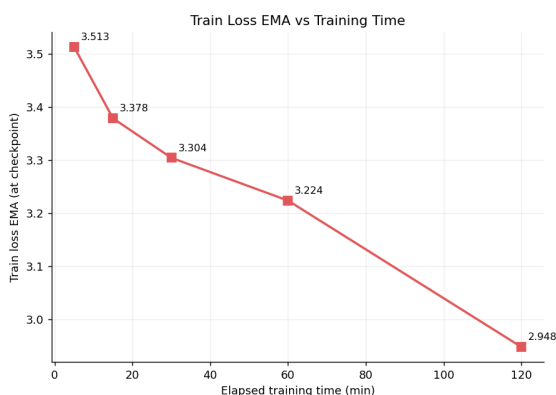
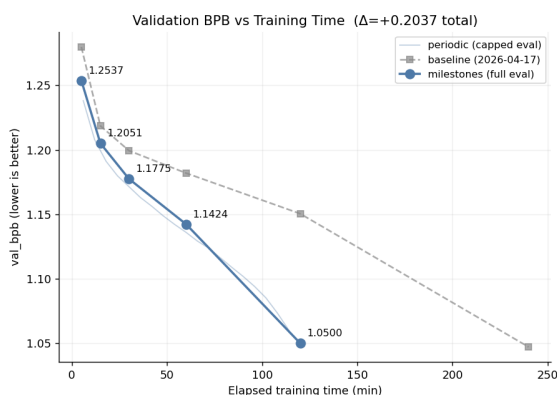
```

        color="#e15759",
        markersize=8,
    )
    for _, row in eval_df.iterrows():
        if pd.notna(row["train_loss_ema"]):
            ax2.annotate(
                f"{row['train_loss_ema']:.3f}",
                (row["elapsed_min"], row["train_loss_ema"]),
                textcoords="offset points",
                xytext=(6, 6),
                fontsize=9,
            )
    ax2.set_xlabel("Elapsed training time (min)")
    ax2.set_ylabel("Train loss EMA (at checkpoint)")
    ax2.set_title("Train Loss EMA vs Training Time")
    ax2.grid(True, alpha=0.2)

plt.tight_layout()
plt.savefig(RUN_DIR / "developmental_curve.png", bbox_inches="tight")
plt.show()
print("Saved output/developmental_curve.png")

```

	label	elapsed_seconds	num_steps	train_loss_ema	val_bpb	elapsed_min
0	5m	300.1	1443	3.512730	1.253732	5.001667
1	15m	900.1	4331	3.378381	1.205067	15.001667
2	30m	1800.1	8664	3.304161	1.177505	30.001667
3	1h	3600.1	17328	3.223759	1.142390	60.001667
4	2h	7200.2	34661	2.948463	1.050018	120.003333



Saved output/developmental_curve.png

1.3 3. Marginal Returns — How Much Does Each Extra Hour Buy?

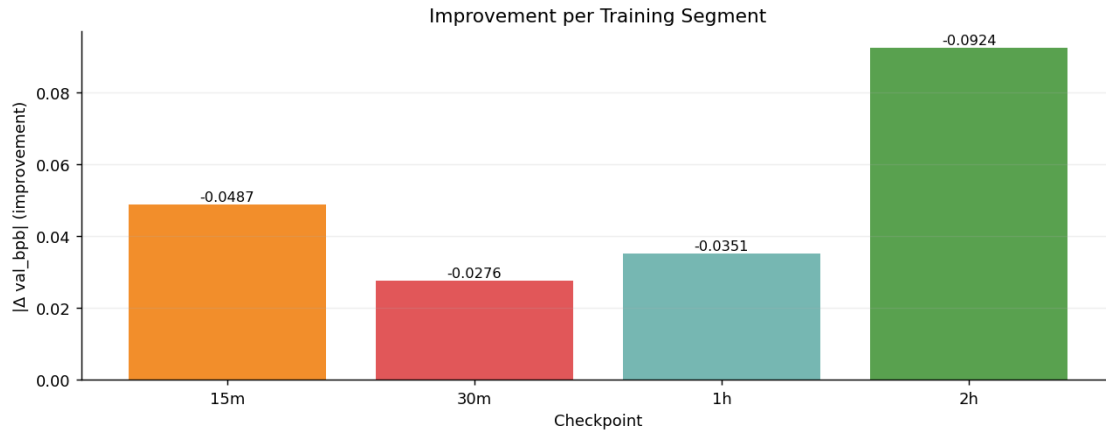
```
[4]: if not eval_df.empty and "val_bpb" in eval_df.columns:
    df = eval_df.dropna(subset=["val_bpb", "elapsed_min"]).copy()
    df["delta_bpb"] = df["val_bpb"].diff() # change from previous checkpoint
    df["delta_min"] = df["elapsed_min"].diff() # minutes between checkpoints
    df["bpb_per_min"] = df["delta_bpb"] / df["delta_min"] # improvement rate

    print("Marginal improvement per checkpoint:")
    print(
        df[["label", "elapsed_min", "val_bpb", "delta_bpb", "bpb_per_min"]].
        to_string(
            index=False
        )
    )

    if len(df) >= 3:
        fig, ax = plt.subplots(figsize=(10, 4))
        valid = df.dropna(subset=["bpb_per_min"])
        bars = ax.bar(
            valid["label"],
            valid["delta_bpb"].abs(),
            color=[COLOR_OF.get(l, "#aaa") for l in valid["label"]],
        )
        ax.set_xlabel("Checkpoint")
        ax.set_ylabel("|Δ val_bpb| (improvement)")
        ax.set_title("Improvement per Training Segment")
        ax.grid(True, alpha=0.2, axis="y")
        for bar, (_, row) in zip(bars, valid.iterrows()):
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.0002,
                f"{row['delta_bpb']:.4f}",
                ha="center",
                va="bottom",
                fontsize=9,
            )
        plt.tight_layout()
        plt.savefig(RUN_DIR / "marginal_returns.png", bbox_inches="tight")
        plt.show()
```

Marginal improvement per checkpoint:

label	elapsed_min	val_bpb	delta_bpb	bpb_per_min
5m	5.001667	1.253732	NaN	NaN
15m	15.001667	1.205067	-0.048665	-0.004866
30m	30.001667	1.177505	-0.027562	-0.001837
1h	60.001667	1.142390	-0.035115	-0.001171
2h	120.003333	1.050018	-0.092372	-0.001539



1.4 4. LR Schedule Shape at Each Duration

Shows how the warmdown curve changes shape as total training time increases, using the current WARMDOWN_RATIO.

```
[5]: # Read current hyperparams from any available JSON sidecar
warmdown_ratio = 0.70
final_lr_frac = 0.05
for label in LABELS:
    meta_path = RUN_DIR / f"model_{label}.json"
    if meta_path.exists():
        m = json.loads(meta_path.read_text())
        hp = m.get("hyperparams", {})
        warmdown_ratio = hp.get("WARMDOWN_RATIO", warmdown_ratio)
        final_lr_frac = hp.get("FINAL_LR_FRAC", final_lr_frac)
        break

def lr_schedule(progress, warmdown_ratio=0.70, final_lr_frac=0.05,
    warmup_ratio=0.0):
    if progress < warmup_ratio:
        return progress / warmup_ratio if warmup_ratio > 0 else 1.0
    elif progress < 1.0 - warmdown_ratio:
        return 1.0
    else:
        cooldown = (1.0 - progress) / warmdown_ratio
        return cooldown + (1 - cooldown) * final_lr_frac

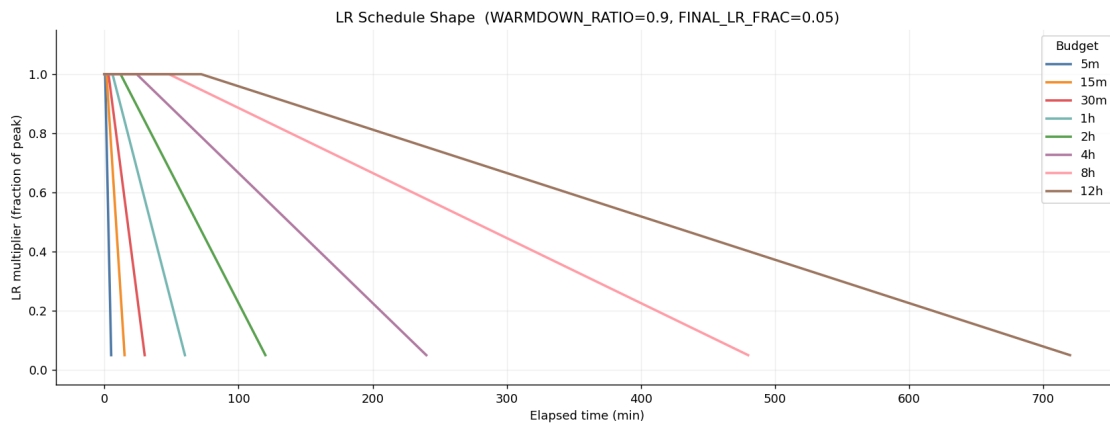
fig, ax = plt.subplots(figsize=(13, 5))
for label, color in zip(LABELS, COLORS):
    budget_s = LABEL_S[label]
```

```

budget_m = budget_s / 60
t = np.linspace(0, budget_m, 500)
progress = t / budget_m
lr = [lr_schedule(p, warmdown_ratio, final_lr_frac) for p in progress]
ax.plot(t, lr, label=label, color=color, linewidth=2)

ax.set_xlabel("Elapsed time (min)")
ax.set_ylabel("LR multiplier (fraction of peak)")
ax.set_title(
    f"LR Schedule Shape (WARMDOWN_RATIO={warmdown_ratio}, \u2192FINAL_LR_FRAC={final_lr_frac})"
)
ax.legend(title="Budget", loc="upper right")
ax.grid(True, alpha=0.2)
ax.set_ylim(-0.05, 1.15)
plt.tight_layout()
plt.savefig(RUN_DIR / "lr_schedule.png", bbox_inches="tight")
plt.show()

```



1.5 5. Prompt Pack Outputs — Side-by-Side Comparison

```

[6]: PROMPT_NAMES = [
    "plain_continuation",
    "factual_fragment",
    "longitudinal_anchor",
    "structurally_awkward",
    "anomaly_lure",
    "signature",
    "continuation",
]

```

```

def parse_prompt_file(path):
    """Parse output/<label>_prompts.txt into a dict {prompt_name:␣
    ↪completion_text}."""
    text = Path(path).read_text(encoding="utf-8")
    blocks = re.split(r"\n\[", text)
    results = {}
    for block in blocks[1:]:
        name_end = block.index("]")
        name = block[:name_end]
        body = block[name_end + 1 :]
        comp_match = re.search(r"COMPLETION:\s*(.*)", body, re.DOTALL)
        if comp_match:
            results[name] = comp_match.group(1).strip()
    return results

# Load all available prompt files
prompt_data = {} # label -> {name -> text}
for label in LABELS:
    p = RUN_DIR / f"{label}_prompts.txt"
    if p.exists():
        try:
            prompt_data[label] = parse_prompt_file(p)
        except Exception as e:
            print(f"[{label}] could not parse: {e}")

print(f"Loaded prompt outputs for: {list(prompt_data.keys())}")

```

Loaded prompt outputs for: ['5m', '15m', '30m', '1h', '2h']

```

[7]: # Display side-by-side for each prompt type
available_labels = [l for l in LABELS if l in prompt_data]

for pname in PROMPT_NAMES:
    md = f"### `{pname}`\n\n"
    has_any = False
    for label in available_labels:
        text = prompt_data[label].get(pname, "(not found)")
        # Truncate for readability
        if len(text) > 600:
            text = text[:597] + "..."
        md += f"**{label}**\n> {text}\n\n"
        has_any = True
    if has_any:
        display(Markdown(md + "---"))

```

1.5.1 plain_continuation

5m > The old man walked slowly toward the river and has never been a difficult time driving by humans, but with a small range of human species and having the perfect balance in mind is a difficult time driving into the lives of a human. If we keep ourselves in check here, the only time you feel like we were here is when there is a whole process of uncheasing our brakes into our own city and here in the Greek world. The most important thing I am going to do is not to talk to them in any way. You want your brakes to be better then you will get brakes. My brakes are better than us for this to tell ...

15m > The old man walked slowly toward the river and he was torn while still watching Dubai's break downward, watching the water and drinking in the rain. Already in the world, Dubai would not have had the chance to reach you because he couldn't be going through any other Dubai jaura. Dubai's breakdown at Peterot, and why Dubai said so in the world.

Question: What was the initial event of the trip? Answer: Peterot took over at Peterot, and why it did not have a chance to reach you because they did the trip?<|reserved_0|>If you had a 200 pic you just kept a tiny miniature miniature. You can fly o...

30m > The old man walked slowly toward the river and flowing out for you in some very little bundles. You get you to think again, as soon as you turn to the river. As the river continues to climb up, it's possible to think again. It's not a fair idea to think that you've had a huge problem-solving for the river but the fact that you think of a river by hand, and is that in fact an easier way to see the people in some very close and clear waters as a side effect of what it is? It could take a 24 hour for a man, or someone has a little background in the construction of the river. Good idea; but don...

1h > The old man walked slowly toward the river and drafted the river, the ancient tree loot, tower, tower, the river was as thick as it was in the loot, tower, or water. "Why does the ancient man walked slow and in some places?" It's not enough as a looter tower if it is buried in ancient lore. "The man walked slow and in some places it has been buried in the middle, but nevertheless it had its own, but it was a long way for you, to show up in this bible, it can be buried in ancient lore. There was some land along the Gulf of Stark where the great looter became the ones that had fallen past the...

2h > The old man walked slowly toward the river and the man walked fast. It would have been quite difficult due to time when the man moved there and would walk almost instantly and take more than an hour of walk and the man was not an effective man to man. The man was the one man walked fast. I'm not worried, my man walking slowly and it looked like. Now I'm the old man walked, and I'm very careful when my man step back and my man walked. To me this might be fun to tell your man how long after you walk, or do you have one man walking fast on the other side of it as long as the man walking slowly...

1.5.2 factual_fragment

5m > The capital of France is Paris, and the population of our chief, it is in danger to our owners and governments. It is also referred to as it is a group of people. The capital of France is defined as the first. It is formed by the name of the group of the capital of America. It is formed by the "game to be." The capital of Britain is formed by the 1910-1980s which the word "game" is actually

“turned up” in the English language. The word translates to “long-legged capital” on English. It is originated in Tennon, France and the term “game to be”Terrini,” which is first defined in the English...

15m > The capital of France is Paris, and the population of France, France, France and other other European countries, The Netherlands, Europe, Spain and Spain, the capital of South and South Southeast and the Czech. Maintaining in the European Paris Against Eruption The European Paris Against Eruption (EPAE) is a European agnap, and also by far the European Paris Against Eruption. European Paris Against Eruption. The Ecological Commission (EPAEC) shows no significant increase in the Eruption of European Paris Against Eruption. This European settled France in her European Paris Against European P...

30m > The capital of France is Paris, and the population of 150 has dropped to approximately 200 to 200. These 3rd-years range from 2000 to 2000, according to the population estimation of the economy. What Are Policy Charseness (PRC)? PRC and Paris Charseness, is a scientific and scientific literature on their subject and may not be perceived, according to the National Institute of Public Health, France.PRCs have been studied in two major categories:PRC and PRC. In this category, this is a scientific notation.PRICE SHIOLE PRC, the Crisphere of the Universe, is the world’s biggest global force in ...

1h > The capital of France is Paris, and the population of Paris is increasing. This year, the city was the first to be able to afford to acknowledge more than 6,600 people of Paris from around the world and to the country in the form of its own Paris, there. The city’s first Paris project is called “Poiro de Newt,” the public will likely give the poi-ro- tobacco, as well as the country’s largest private, public and public fundraisons. Still, the urban infrastructure cost would be inadequate, the city is still in poorence and the country in Paris from the city has reached public fundraisons by th...

2h > The capital of France is Paris, and the population of the Parisians is in Europe. Most of this, but the country is in France’s Land The Rise of Non-Korbigmans: Land as well as some of them cancel out. They can’t be bought for sale. They can’t be handed to other country I was reading that one last thing about Paris and this is because I was very familiar with the idea of how the local government is able to stop them from getting there. There are many governments, some of these governments that stop their politics from being able to stop the people (even just that the political issues like th...

1.5.3 longitudinal_anchor

5m > In one sentence, the meaning of life is a term for the “life,” refers to how a person has a connection to one when a person who has a negative impact in life is the life of someone’s most important life. When the person comes to life, or person’s life experiences, often experience a negative impact on one’s life. The last story is told by a man who has 6.15 people. When the person wants to see the life of life, the person he learies it, or does it feel a lost life. He is a personal and social, person, and a life acting on that thing. He is a personal, social and economic person. Second book...

15m > In one sentence, the meaning of life is lost in life and death by the beginning of a century, so the meaning of life is lost by the beginning of the year.

Looking at, there is a need to be in the life of the people in the world. In that way, people are doing a work, and they are doing a role (like a person sitting or watching TV) and simply sharing their opinions with their friends and family; and perhaps talking about people from different things. A person who has a life can be a person (humanity) that feels like a person, or is like a single person but who makes your work, work (these pe...

30m > In one sentence, the meaning of life is that of a stone, another, or a quart, are actually a great example of someone else's strength. One piece of the stone is from the road it is a road where you get a chance of something. In a stone, it means you get a chance of something that someone thinks is it that you have an unprecedented relationship for which you've started it for just about 200 years and even 130 years old. If the stone has an unprecedented relationship (like stone). By that time you can ask that even more for the rest? Why is it possible to be a stone a piece of the road it use...

1h > In one sentence, the meaning of life is intermined by those words. The reality is that life is connected to the soul, a person's stenciliating feeling so that they cannot exist, so they can, hence, be called life. For example, the life is divided into two groups: the same thing or the same as the same thing or the different thing: the same thing, and the opposite thing: the same in different ways. But things happen more and more so in the same way.

Of course, these same things are called life.

It is a term which refers to those of us who are wealth or who are just living in different ways...

2h > In one sentence, the meaning of life is based on a particular meta-analysis. While it may seem to be similarly obvious, life is still too harsh to develop as you make progress. Many people who believe life is the most important thing. When you are thinking of living a long life, then life isn't always an answer to the question: Why is life and life important? " " " Life as well as life is considered the most significant part of the global population. And life as well as life itself, there is much to learn, as well as a skill – as well as the ability to understand what things are. " " " ...

1.5.4 structurally_awkward

5m > Despite the fact that however, the reason why because the fact that I actually told my dog is really wrong, isn't the only reason actually? We're actually the actual question of what's happening. We know when you read, if you read, the answer is your question with my answer. This is right, you will probably be the only question that I got the right question. That's not how science plays in mystical role, but also the fact that I have said to be a big reason for those who are trying to stand up, there is no way for me because I won't come away due to many of us, as we are now going to come a...

15m > Despite the fact that however, the reason why because different colors are the size of the shell, most commonly called coloring, are not too smaller (mature, not too fussly or too large). A smaller size of the shell means a 55x25x diameter. This means the shell must not be smaller. Because the larger shell the size of the shell, these large shells, called lilac shells, can take 12 to 24 mm, so the size is 2. International Lily and Cockage It is not just a large size shell. The shell must not be in a typical range of sizes, but also includes about 55x25 larger shells. Weight, large shells ca...

30m > Despite the fact that however, the reason why because of the problem is because of its

obvious limitations as well. The solution is to determine why the solution does not matter where the reason is that while the solution may have to be replaced by the solution. For example, for 9 years 2000 a while is probably 2 years old, or 3 months old if 1 years old, however, there are so many advantages to the solution. When was the solutions? You know they might be 3-4 years old.

Disadvantages

Good quality in its solution is not the reason. In order to determine why. First off, we do not need to lo...

1h > Despite the fact that however, the reason why because it's true that the most important rule in the world is that if you have at least four tolls in the universe (or rather the way they could) you get to know it at least, to know that it's true that there is no rule or even no rule that in a number sense, there are no rule that in the universe. If it doesn't know or even just know that there are no rule, the truth about it is true but not even there. That is why it really matters most not because of no rule you can understand is true but even if you're not certain people are trying to disco...

2h > Despite the fact that however, the reason why because of their poor eyesight. They never thought of the fact that we were all just like people, we could see some more about how we have come to believe in what is happening, and how and why it goes.

When I look back at the original question I came across, people have been saying "toucan" to me and I think "toucan" to go. When it comes to that question. While people have been saying they do nothing at all! Even though it's the fact that it's a fact, "touche" is a question of sorts and not a fact, but in many cases it becomes a fact and its a ...

1.5.5 anomaly_lure

5m > Ground control to Major Snorf, Cakery, Weather Rabbits This beautifully reminds you of sageways to wrap into their mouths, not just not to do so, and that they actively support them. For most children, sageways, including adults, cats, cats, cats, cats, and some of their territorial nature. This is a particularly true source of connection to catacre and the importance of active nature. Fortunately, adult cats are naturally echocrats, that their food is often raised as predators. Rabbits, including humans, are protected from pets from the environment and will develop their nuance. These cats...

15m > Ground control to Major Snorf, Leechi, Rick, Shirmer Ground control to Major snakes Ground control should go on, but a snake should not be a very tallied snake's shoulders. You can also grow a snake's in some cases, while some snakes might be a small snake's shoulders. For example, the snake should have a strong curd shape, and in some cases, it is essential to provide a snake with a balanced curd shape. To propagate through ground control should go through a snake-about. The snake should be tightly spatial to the base, between the shoulder, back and knee to the base with a horizontal elbow...

30m > Ground control to Major Snorf, or a member of the masquis are common among allied Mother Nature and Division of Dahara Mandison's Mother Nature. If you have any questions on why you decide to go for an Mother Nature it's obvious that you have a choice between the species, the plants, or the flowers in this area and its habitats (even ground areas)—like birds in the Mother Nature. While it's possible, a carnival of other Northwest Mother Nature's native plants, all their

species like deer, praise, cucurus, melons! The plant hosts insects at the surface of the leaf litter on the top of a gras...

1h > Ground control to Major Snorf, Friendslit Mur. Sullivan. Ground control and home management. Uptake of earthquake control is a basic obligation for all civilians in the U.S., but no one knows. The rule is to get rid of all of these variables to the earthquake control of Mongolia, which includes more than 100,000 tonnes or about five percent of the total population of Mongolia. The rule is to obtain a ground control for Mongolia, as well as the earthquake control for Mongolians, such as the Mongolians and Sussex. So, to the extent that each Mongolian is different, it is a separate control of...

2h > Ground control to Major Snorf, with its latest

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Ground-balancing is an intriguing way of enhancing safety features in an upcoming, ever-loving NPR Senator.

NPR Sensing

Ground-balancing is the method of building smart, safe, comfortable clothing that has undergone advanced safety testing. Eco-friendly technology with its exceptional weather-fighting capabilities and advanced features make this innovative approach a must-have for every gamer. From safety features and accessories to its upgraded functionality and security features, this platform offers an extensive rang...

1.5.6 signature

5m > Once upon a time there was a small town among the residents, town, and tie, there lived a long range of about a 50s and two. Robot (5) Dolphins, which lived at the 20th annual in Swiss. Unsynopsy. It seemed to have a very high visibility. It is a young houseplant. With friends being with friends or family and you have fun, a person could be on their own. Cost About: 10th January With the exceptional bird, there are 50 deaths for birds, including: Today's birds. Many birds consume food. For more views, the bird is usually seen in birds. Dolphins. They eat food, as birds consume food. The bir...

15m > Once upon a time there was a small population of Grayolaasaaaaaaaaa, he realized the idea is that they didn't do this. This "bad" is not a bad thing, either! However, as we may see the gay, we will have to go into something other than to put us on a table. One time he could see the gay, the smaller ones on the table. In our practice, we can study it, which shows it back into its habitual habit. Now here, what does a superb or superconducting gay? The gay, we can expect one or the others to become too close to another person in his hand. So imagine, we wouldn't say they was just another indi...

30m > Once upon a time there was a small town filled with airsoftware, there decided to start the Fabi, where the first thing looked and explained which things were different. This allowed me to join a conjunctiva (platinum, bolderos) where you wanted to share a real niche called a "citizative" or "picture."

On the two sides – Fabi was called a pintel, who was about two weeks later than Faul (they kept being on the right but he had to administer to a different representation). Sometimes the most

Persian, who took a family traded the Persians to the Persian, reached Europe by 0.5 million years of age. It was founded by John H. Baptism, who the Bible was among the biggest German citizens in the 1840s, ...

2h > The instructions were clear until line seven: A. Dungan

B. Gerald F. Crick

C. Ford

Answer: B. Dungan

D. Lloyd

D. Jamie G. Rick

Answer: C

Lewis Armit

Answer: G Rick

D. Ford

Answer: B. Dungan

Answer: C

H

Lewis Armit

Answer: B

Lewis Armit

Answer: L

Hewis Armit

Answer: H

Lewis Armit

Answer: H

Hewis Armit

Answer: H

Lewis Armit

Answer: B

Lewis Armit

Answer: H

Lewis Armit

Answer: H

Lewis Armit

Answer: H

Hewis Armit

Answer: H

Lewis Armit

Answer: Her

Her

Hewitzing

questions

=====

1.6 6. Text Quality Metrics

Automatic proxies for qualitative change. These are rough signals, not ground truth.

```
[8]: def repetition_onset(text, ngram=4):
    """Word index of the first repeated n-gram. Returns None if no repetition.
    ↪"""
    words = text.lower().split()
    seen = {}
    for i in range(len(words) - ngram + 1):
        g = tuple(words[i : i + ngram])
        if g in seen:
            return i
        seen[g] = i
    return None

def longest_clean_span(text, ngram=4):
    """Number of words before the first repeated n-gram (= repetition_onset, or_
    ↪full length)."""
    onset = repetition_onset(text, ngram)
    words = text.split()
    return onset if onset is not None else len(words)

def repetition_rate(text, ngram=4):
    """Fraction of n-grams that are repeated."""
    words = text.lower().split()
    if len(words) < ngram + 1:
        return 0.0
    grams = [tuple(words[i : i + ngram]) for i in range(len(words) - ngram + 1)]
    return 1 - len(set(grams)) / len(grams)
```

```

def sentence_completion_rate(text):
    """Fraction of sentence-like units ending with . ! ? - proxy for syntactic
    ↪tidiness."""
    units = re.split(r"(?<=[.!?])\s+", text.strip())
    if not units:
        return 0.0
    complete = sum(1 for u in units if re.search(r"[.!?]", u.strip()))
    return complete / len(units)

def type_token_ratio(text):
    """Lexical diversity - length-dependent; use as directional signal only."""
    words = re.findall(r"\b\w+\b", text.lower())
    return len(set(words)) / len(words) if words else 0.0

def avg_sentence_length(text):
    sentences = re.split(r"[.!?]+", text)
    lengths = [len(s.split()) for s in sentences if s.strip()]
    return np.mean(lengths) if lengths else 0.0

# Compute metrics per label per prompt
metric_rows = []
for label in available_labels:
    for pname, text in prompt_data[label].items():
        onset = repetition_onset(text)
        metric_rows.append(
            {
                "label": label,
                "prompt": pname,
                "rep_onset_word": onset if onset is not None else len(text.
                ↪split()),
                "clean_span": longest_clean_span(text),
                "rep_rate": round(repetition_rate(text), 4),
                "sent_completion": round(sentence_completion_rate(text), 3),
                "ttr": round(type_token_ratio(text), 4),
                "avg_sent_len": round(avg_sentence_length(text), 1),
            }
        )

if metric_rows:
    mdf = pd.DataFrame(metric_rows)
    summary = (
        mdf.groupby("label") [

```

```

        [
            "rep_onset_word",
            "clean_span",
            "rep_rate",
            "sent_completion",
            "ttr",
            "avg_sent_len",
        ]
    ]
    .mean()
    .round(3)
)
summary = summary.reindex([l for l in LABELS if l in summary.index])
print("Mean across all prompts per checkpoint:")
display(summary)

# Plot: the four most diagnostic signals
fig, axes = plt.subplots(2, 2, figsize=(14, 8))
plots = [
    ("clean_span", "Longest clean span (words)", "#4e79a7", "higher is_
↪better"),
    (
        "rep_onset_word",
        "Repetition onset (word index)",
        "#76b7b2",
        "higher is better",
    ),
    ("rep_rate", "Repetition rate (4-gram)", "#e15759", "lower is better"),
    ("sent_completion", "Sentence completion rate", "#59a14f", "higher is_
↪better"),
]
for ax, (col, title, color, note) in zip(axes.flat, plots):
    ax.plot(
        summary.index,
        summary[col],
        marker="o",
        color=color,
        linewidth=2,
        markersize=8,
    )
    for x, y in zip(summary.index, summary[col]):
        ax.annotate(
            f"{y:.2f}",
            (x, y),
            textcoords="offset points",
            xytext=(0, 8),
            ha="center",

```

```

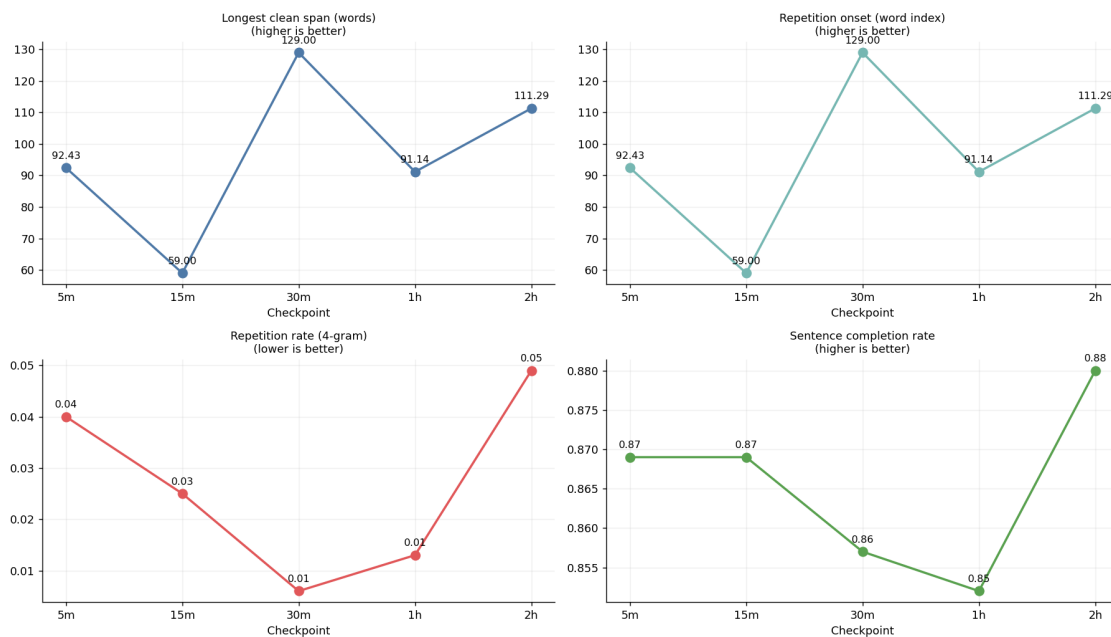
        fontsize=9,
    )
    ax.set_title(f"{title}\n({note})", fontsize=10)
    ax.set_xlabel("Checkpoint")
    ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.savefig(RUN_DIR / "text_metrics.png", bbox_inches="tight")
plt.show()
print("Saved output/text_metrics.png")

```

Mean across all prompts per checkpoint:

	rep_onset_word	clean_span	rep_rate	sent_completion	ttr \
label					
5m	92.429	92.429	0.040	0.869	0.509
15m	59.000	59.000	0.025	0.869	0.552
30m	129.000	129.000	0.006	0.857	0.510
1h	91.143	91.143	0.013	0.852	0.511
2h	111.286	111.286	0.049	0.880	0.532

	avg_sent_len
label	
5m	19.143
15m	16.486
30m	20.629
1h	20.329
2h	18.214



Saved output/text_metrics.png

1.7 7. Qualitative Notes

Per-run scoring tables and narrative observations now live next to the run data, in `<run_dir>/qual-notes.md`. The rolling cross-run analysis (plateau evidence, warmdown findings, decisions for the next night) stays in [docs/phase2-developmental-study.md](#). Memorable generations are collected in [docs/curios.md](#).

The current run's `qual-notes.md` is rendered below; if `BASELINE_DIR` has one, it is shown too for side-by-side comparison. If a run has no `qual-notes.md` yet, the cell prints a short hint instead.

```
[9]: def _show_qual_notes(run_path, heading):
      notes = Path(run_path) / "qual-notes.md"
      if not notes.is_file():
          display(Markdown(f"_No `qual-notes.md` in `{run_path.name}` yet._"))
          return
      display(Markdown(f"### {heading} - `{run_path.name}`"))
      display(Markdown(notes.read_text(encoding="utf-8")))

      _show_qual_notes(RUN_DIR, "Current run")
      if BASELINE_DIR is not None:
          _show_qual_notes(BASELINE_DIR, "Baseline")
```

No qual-notes.md in 2026-04-19_2h_warmdown_0.90_run yet.

No qual-notes.md in 2026-04-17_phase2_baseline yet.